



Why are Europe's underserved still unprotected? A call to address structural barriers to immunisation

Alison Maassen, Afsaneh Nejat, Janine de Zeeuw, Lara Tavoschi, Pania Karnaki, Daniela Filáková Bobáková, Maria Elena Tosti, Maurizio Marceca, Erica De Vita, Irina Barbîroş, Michael Edelstein

Persistent immunisation inequities hinder progress towards public health goals across the WHO European region. In this Health Policy, we synthesise evidence from three EU-funded projects—RIVER-EU, AcToVax4NAM, and RISE-Vac—to identify key structural barriers and enabling factors that affect vaccine uptake among underserved populations, including people of minoritised ethnicity, newly arrived migrants, and people living in prison. Through a health-systems lens, we present actionable, system-level policy recommendations aligned with current EU and WHO strategies. These recommendations include enhancing service integration, prioritising inclusive governance, strengthening data systems, and addressing underinvestment in vaccine delivery. Our findings are directly relevant to the implementation of the European Immunization Agenda 2030, health-related conclusions of the EU Council (2024, 2018), and Principle 16 of the European Pillar of Social Rights. By addressing structural determinants, our recommendations offer opportunities to improve vaccination uptake among both underserved groups and the general population. We aim to contribute timely evidence to support health-system reforms that build trust, resilience, and equity across Europe.

Introduction

Tackling low vaccine coverage is a public health priority, as vaccines are among the most cost-effective and successful public health interventions.¹ Yet, vaccine coverage has stalled or declined across parts of the WHO European region.² Too often, coverage is piecemeal, with substantial gaps within and between countries and, oftentimes, experienced by specific populations.³

The causes of under-immunisation in these specific populations are complex and multiple. Labels such as hard-to-reach individuals or populations could imply reticence or unwillingness to engage with immunisation programmes, whereas, in reality, long-standing structural inequalities have a major influence on the access of these populations to health-care services in general and immunisation services in particular. Vaccine hesitancy is often cited as the primary cause of under-immunisation, but is rarely the sole or dominant factor.^{4–6} A conceptual alternative to the hard-to-reach concept is framing these groups as underserved populations: groups of individuals that do not have high vaccine coverage within their local immunisation programmes. Socioeconomically disadvantaged populations; minoritised cultural or religious groups; people living in medically underserved areas; and those who are otherwise isolated from the wider population, such as people living in prison and newly arrived migrants (including those residing in asylum centres) often fall into this category. Unlike the term hard-to-reach, the term underserved puts the onus on the health system to deliver routine vaccination programmes in ways that are equitable and account for intersectional inequalities of race, ethnicity, socioeconomic status, and gender, among other characteristics.⁷ In this Health Policy, we define populations falling into these categories as underserved populations, who often have substantially lower vaccine uptake than the general population.⁸

Despite scarce data on vaccine coverage in underserved populations, substantial differences in vaccine coverage rates are reported. Across Europe, vaccination coverage among people living in prison is largely unmonitored; however, from available data, presented in the status report for 2022 by WHO, only 17 (50%) of 34 countries offered vaccination against human papillomavirus (HPV) in prisons and only 24 (66.7%) of 36 countries offered vaccination against hepatitis B virus.⁹ In Greece, the uptake for second dose measles-containing vaccines in 2019 was only 45% in migrant children aged 0–4 years.¹⁰ In Slovakia, in 2022, the regions with the highest Roma populations¹¹ show the lowest HPV vaccination coverage.¹²

Although extensive work has explored attitudes and beliefs about vaccination, there is not enough research into systemic barriers and enablers for underserved populations.¹³ Available evidence often overlooks structural barriers and focuses on perceived population-side barriers, that is, vaccine decision-making by individuals and communities.¹⁴ In this Health Policy, we aim to address this gap, drawing on findings from three EU-funded projects—RIVER-EU, AcToVax4NAM, and RISE-Vac—that explore system-level barriers using innovative and participatory methods to advance vaccine equity (table).

Structural barriers and enabling factors via the WHO Health System Building Blocks framework

For this Health Policy, we conducted a comprehensive review of original research, project deliverables, and reviews carried out within the three EU-funded projects.

For RIVER-EU, all peer-reviewed publications were reviewed alongside two submitted but non-public deliverables, which were included following consultation with the respective authors to ensure a complete representation of the findings. For AcToVax4NAM, given that major findings were not fully reflected in the project's peer-reviewed

Lancet Prim Care 2025;
1: 100062

Published Online November 20,
2025
<https://doi.org/10.1016/j.lanprc.2025.100062>

EuroHealthNet, Brussels, Belgium (A Maassen MSc, A Nejat MD); Global Health Unit, Department of Health Sciences, University Medical Center Groningen, Groningen, Netherlands (J de Zeeuw PhD); Department of Translational Research and New Technologies in Medicine, University of Pisa, Pisa, Italy (Prof L Tavoschi PhD, E De Vita PhD); Prolepsis Institute, Athens, Greece (P Karnaki MSc); Faculty of Medicine, PJ Šafárik University, Kosice, Slovakia (D Filáková Bobáková PhD); Social Health Institute, Palacký University Olomouc, Olomouc, Czech Republic (D Filáková Bobáková PhD); Centro Nazionale per la Salute Globale, Istituto Superiore di Sanità, Rome, Italy (M E Tosti MSc); Department of Public Health and Infectious Diseases, Sapienza University of Rome, Rome, Italy (M Marceca MPH); Medical Department, National Administration of Penitentiaries Moldova, Chisnau, Moldova (I Barbîroş PhD); Azrieli Faculty of Medicine, Bar-Ilan University, Safed, Israel (Prof M Edelstein MbChB)

Correspondence to:
Ms Alison Maassen,
EuroHealthNet, Brussels 1000,
Belgium
a.maassen@eurohealthnet.eu

	Scope	Target beneficiaries
RIVER-EU (Reducing Inequalities in Vaccine Uptake in the European Region—Engaging Underserved Communities, 2021–26) ¹⁵	Addressing health system barriers in measles, mumps, and rubella and human papillomavirus vaccination for different underserved populations	Marginalised Roma communities in Slovakia, Ukrainian migrants and refugees in Poland, migrants and refugees in Greece, and adolescents with Turkish and Moroccan migration backgrounds in the Netherlands; empowering examples include the Somali community in Finland, the Arab community in Israel, and the Bangladeshi community in the UK
RISE-Vac (Reaching the hard-to-reach: increasing access and vaccine uptake among the prison population in Europe, 2021–24) ¹⁶	Promoting vaccine literacy, enhancing vaccine offering, and increasing vaccine uptake for people who live and work in prisons	People who live and work in prisons across Cyprus, France, Germany, Italy, Moldova, and the UK
AcToVax4NAM (Access to Vaccination for Newly Arrived Migrants, 2021–24) ¹⁷	Improving vaccine literacy and access for newly arrived migrants	Newly arrived migrants across Cyprus, Germany, Greece, Italy, Malta, Poland, Romania, and Spain

Table: Summary of the RIVER-EU, RISE-Vac, and AcToVax4NAM projects

literature, additional information was gathered from publicly available project deliverables. For RISE-Vac, all relevant peer-reviewed articles, along with the project's landmark policy brief, were included. For RIVER-EU, when content appeared in both public and restricted-access formats, priority was given to citing publicly available sources, except in one instance in which unpublished but submitted findings from Greece were used due to the absence of a publicly available equivalent.

Our analysis excluded findings related to behavioural, attitudinal, and individual-level factors related to vaccine hesitancy or personal decision making, as our focus was on structural and system-level determinants. Therefore, only findings that could be clearly mapped to structural or systemic determinants within the WHO Health System Building Blocks framework were included.¹⁸

This WHO framework was used due to its methodical approach for examining service delivery, workforce, financing, information systems, essential medicines, and governance.¹⁸ Although the framework does not capture underlying social determinants or individual decision making, it incorporates key elements for addressing health inequalities: availability, accessibility, affordability, and quality of services. Based on the nature of the findings that emerged from the included materials, we made targeted adaptations to the framework, to better reflect the realities of vaccine delivery for underserved populations.

These adaptations were made because many of the barriers and enablers identified in the project data extended beyond the formal health system and involved sectors such as the judicial system, education, migration, and social care. In line with this observation, the findings showed that effective vaccination strategies require a broad range of actors, often outside traditional health professionals, and that equity in immunisation access depends on a human rights-based and life-course approaches, rather than simply ensuring availability of vaccines within clinics.

Specifically, the following three adaptations were made. First, the term health workforce was reframed as professionals for health (PFH), a term introduced by the AcToVax4NAM project. This broader concept includes all health-care and social-care professionals involved in vaccination services at local, regional, and national levels. The concept covers not only those directly delivering vaccines

(eg, nurses and general practitioners) but also those involved in organising, coordinating, or facilitating access (eg, cultural mediators, school staff, interpreters, and planners). Second, the term essential medicines, which originally focused on the availability and affordability of essential medicines, was adapted to the term access to essential immunisation. This interpretation allowed us to focus on vaccine-specific considerations, such as the right to life-course vaccination, cold-chain functionality, and supply-chain reliability. These elements are central to immunisation equity but less visible in the original medicines-focused block. Third, the term leadership and governance was maintained but its interpretation expanded from establishing policy frameworks to reflect cross-sectoral coordination and institutional trust.

Barriers and enablers that emerged from our analysis were categorised within each revised building block (figure) and grouped into relevant subcategories. This ordered but adaptive approach enabled a comprehensive understanding of systemic factors affecting vaccine uptake and offered a health systems-oriented lens to assess and prioritise actions to improve vaccine equity across Europe. Acting on these factors can benefit multiple underserved populations and enhance overall health-system equity, trust, and value for money.

Systemic barriers to and enablers of access to immunisation services for underserved populations

Building block 1: service delivery

Access and clinic hours

Convenient locations, welcoming settings, and flexible hours contribute to easier vaccine uptake, whereas distant clinics, rigid appointment systems, and complex procedures can create barriers—especially for rural residents or those with low digital access.^{14,19,20} Although mobile units can temporarily improve access, integrating clear, accessible pathways to vaccination into existing health services is deemed more sustainable.¹⁹

Outreach and communication

Health services often do not have sufficient proactive outreach or counter-narratives to anti-vaccination messaging.¹⁴ During the COVID-19 pandemic, HPV campaigns lost

	Building block components	Barriers	Enablers
Building block 1: service delivery Delivering safe, effective, and quality immunisation services (including awareness-raising campaigns, appointment bookings, and vaccinations) to all who need them, when and where they are needed	1.1 Access and clinic hours	Inconvenient clinic locations, hours, and rigid processes (eg, distant clinics, inflexible opening hours, complex prescription-based processes, and digital-only bookings)	Vaccination settings are made more accessible (eg, welcoming environments, convenient locations, mobile units, and longer hours)
	1.2 Outreach and communication approaches	Passive and unresponsive systems (eg, non-existent or inappropriate outreach, not addressing anti-vaccination messaging, and not actively promoting the value of vaccines)	Active outreach; information, education, and communication interventions optimally delivered by trusted actors (eg, community members, and people with lived experience)
	1.3 Tailored, multilingual information	Lack of translated and culturally appropriate information materials (eg, confusion between HPV and HIV in materials translated to Romani)	Culturally sensitive communication available in multiple languages (eg, framing HPV vaccination as cancer prevention instead of protection against a sexually transmitted infection)
	1.4 Continuous and integrated services		Continuity of services and timely introduction of vaccination information at all health-care and administrative touchpoints (eg, introducing information during pregnancy and upon admission to prisons or reception centres and requiring vaccination records at school enrolment)
Building block 2: professionals for health Having sufficient well trained and fairly distributed health professionals (eg, doctors and nurses) and auxiliary support (eg, cultural mediators and interpreters)	2.1 Trust in professionals for health		Professionals' active recommendation of vaccines as the most trusted source of information (eg, presence of health-care staff in reception centres and prisons enhances trust) Personal characteristics of professionals: perceived as trustworthy, accessible, and non-discriminatory
	2.2 Capacity of human resources	Shortage of health workers; less time to discuss and educate about vaccines Language and communication barriers; shortage of interpreters and cultural mediators	Good collaboration between health professionals and auxiliary support and task shifting (eg, translators, school staff, and prison custodial staff)
	2.3 Training and confidence in communication and knowledge of rights	Inadequate training on vaccine communication (for health workers and interpreters); workforce not culturally competent Vaccine hesitancy among professionals (eg, advising HPV vaccine as unnecessary on the basis of assumptions about sexual activity) Little professional awareness on rights to access vaccines (eg, for newly arrived migrants)	
	2.4 Cultural and linguistic competence and representation	Untrained interpreters can undermine trust (eg, wrong translations)	Cultural representation and tailored language access (eg, shared cultural background and use of interpreters and cultural mediators)
Building block 3: immunisation information systems Collecting, storing, and managing individual immunisation records that are transferable across services and regions while adhering to data privacy and protection standards; adequate use of records to support follow-up and catch-up campaigns	3.1 Continuity and unification of data systems	Insufficient continuity of records, temporally or across systems (eg, no central immunisation register); loss to follow-up in mobile populations or those with high turnover Paucity of protocols for handling missing records (eg, vaccination records of newly arrived migrants are often paper-based, easily lost, or not integrated into national databases)	
	3.2 Administrative and procedural processes	Inhibiting or cumbersome administrative or technical procedures (eg, identification numbers or translated records for newly arrived migrants)	Effective immunisation recordkeeping (eg, Personal Child Health records)
	3.3 Reminder and recall systems	Financially sustained and tailored reminder systems	Proactive reminder and recall systems relying on unified data systems (eg, automatic SMS reminders)
	3.4 Evaluation and monitoring	Paucity of coordinated evaluation and monitoring systems, disaggregated data (eg, non-governmental organisation-led evaluations that are rarely shared with state services)	Qualitative and inclusive monitoring systems
Building block 4: access to essential immunisation Ensuring people can exercise their right to life-course vaccination and that adequate supply chains and cold chains are in place to maintain availability and quality and reduce waste	4.1 Availability	Life-course vaccination services not routinely available in all settings (eg, prisons and reception centres); outbreak driven and supply driven Shortage of equipment to safely maintain vaccines and reduce wastage (eg, cold chain) Political debates about so-called deservingness (eg, for asylum seekers or people living in prison)	
	4.2 Eligibility	Restricted or prohibited access based on age, gender, or insufficient documentation (eg, HPV vaccines offered only to girls at specific ages)	Universal, equitable, decentralised, and continuous vaccination programmes across generations Expanded access to specific vaccines for underserved individuals, considering increased risks (eg, gender-neutral HPV vaccination in prisons because of over-representation of groups at high risk)
Building block 5: financing Reducing out-of-pocket payments and insurance loopholes or copayments for individuals to receive recommended vaccines; ensuring adequate health budget for staff and to maintain immunisation services for all	5.1 Direct costs	High out-of-pocket cost (or perceived cost) of vaccination to individuals	Free vaccination programmes that ensure equitable access for low-income and marginalised communities
	5.2 Indirect costs	Inflicted costs such as travel, lost income, childcare, and digital (eg, access to electronic devices and internet)	Reducing indirect costs through local availability of services and travel support to sustain uptake
	5.3 Systems funding	Underfunded vaccination systems with insufficient procurement budgets and cold-chain infrastructure leading to prioritisation of low-cost vaccines over more expensive ones	Adapting vaccination schedules (eg, accelerated hepatitis B or single-dose HPV regimens)
Building block 6: leadership and governance Providing strategic policy frameworks; clear standard operating procedures; and guidance on right to vaccination, individuals with missing vaccination records, and others Facilitating cross-sectoral collaboration and coordination among relevant sectors and stakeholders	6.1 Trust and institutional frameworks	Little trust in health institutions and authorities (eg, distrust towards prison authorities, fear of deportation among migrants, distrust in health institutions after the COVID-19 pandemic)	Trusted and reliable institutions
	6.2 Guidelines and standard operating procedures	National or local policy not explicitly specifying right to vaccination for some communities (eg, refugees, migrants, or people living in prison) Unclear guidelines regarding vaccination in specific circumstances (eg, missing vaccination records, unaccompanied minors)	National and subnational guidelines for underserved populations (eg, people living in prison)
	6.3 Partnerships and integration	Divergent organisational cultures and priorities when vaccine delivery is shared (eg, between Ministry of Justice and Ministry of Health)	Integration of immunisation services into multiple contact points (eg, general practitioner practices, mosques, school-based vaccination delivery, and prisons) Cross-sector partnerships (eg, integration of prison health systems into public health systems, collaborations between state services and civil society organisations for newly arrived migrants)

Figure: Modified WHO building blocks and identified barriers and enablers to vaccine uptake in underserved populations
 HPV=human papillomavirus.

visibility²¹ and often missed target communities as intensifying digitalisation and the absence of translated information created misunderstandings about vaccine eligibility, costs, and schedules.²² Among newly arrived migrants, vaccine campaigns often do not use appropriate communication channels.¹⁹

Tailored, multilingual information

Campaigns might be less effective when materials are overly complex, poorly designed, or untranslated or culturally misaligned.^{14,19,21,22} Roma health mediators noted that not tailoring information for people with lower literacy could lead to confusion between terms such as HPV and HIV.¹⁴ In conservative contexts, framing HPV as cancer prevention can be more effective than messages centred on sexual health.²⁰ Research emphasises the importance of involving individuals with lived experience throughout the research process.²³ In prisons, coproduction of educational materials for people living in prisons helped identify key content and preferred delivery channels.²⁴

Continuous and integrated services

Integrating vaccination into other points of contact with the health system, such as antenatal care, school enrolment, refugee and asylum reception centres, or prison admission, helps to create timely opportunities to inform and vaccinate underserved populations. This opportunistic approach supports continuity of care and reduces missed opportunities.^{19,24–26}

Building block 2: PFH

Trust in PFH

In underserved populations with high vaccination coverage, health-care professionals are the most trusted information source; this belief is consistent with beliefs in the general public.^{20,26,27} Trust can compensate for low health literacy, with decisions based more on confidence in health-care providers than on a full understanding of vaccines.^{26,27} Including doctors and translators in services can enhance trust among newly arrived migrants.^{19,28} In prisons, building trust required health-care staff to operate independently from correctional authorities, with involvement of peer educators or personal networks to boost vaccine confidence.^{29,30} Other intrinsic characteristics of health-care professionals, such as perceived trustworthiness, respectful treatment, and non-discrimination, are considered essential for vaccine uptake.^{14,22,31}

Capacity of human resources

Insufficient human resources risk reducing time and opportunities for tailored recommendations and discussions.^{14,21,22} Overstretched professionals often prioritise immediate care over vaccination conversations.¹⁴ A lack of cultural mediators and interpreters can further limit access to immunisation services for people from diverse backgrounds.¹⁹ In prisons, task shifting to trained non-medical staff, including custodial staff, can help, but the perceived

independence and trustworthiness of the health-care staff should not be undermined in the process.^{32,33}

Training and confidence in communication and knowledge of rights

Many PFH do not have adequate training in effective vaccination communication and feel unprepared and hesitant to discuss it.^{19,31} This lack of training is common with respect to vaccines linked to sexual health. In the Netherlands, fear of backlash made some professionals avoid conversations about the HPV vaccine.²¹ In Greece, cultural taboos around discussing sexuality led to some health professionals not informing migrant families about the HPV vaccine, despite high interest among mothers and daughters (unpublished, RIVER-EU). In the Netherlands, some professionals reinforced misconceptions by advising that HPV vaccination was unnecessary for migrant women who were not sexually active before marriage.²¹ Furthermore, technical language and little awareness of migrant rights—specifically, whether recently arrived migrants are entitled to receive vaccination—can cause missed opportunities.^{14,19} Panel 1 presents a case study that highlights the role of PFH.

Cultural or linguistic competence and representation

People feel more comfortable when health-care providers share their background, speak their language, or understand their norms.^{20,27} Although system-wide cultural matching does not seem feasible given the diversity of patient populations and workforce constraints, interpreters and cultural mediators can improve communication and care quality.^{19,26} Verbal explanations in a person's native language are generally preferred over written materials because individuals are more likely to listen to information than read materials.^{20,21,27} However, untrained interpreters can reduce clarity and trust if their personal beliefs influence translation.¹⁹

Building block 3: immunisation information systems

Continuity and unification of data systems

Fragmented data systems mean vaccination records are often unable to follow individuals across settings or regions. In prisons, inadequate linkages between prison health services and community health services, combined with the high turnover, results in incomplete multidose vaccine courses.^{32,33} Similar gaps affect newly arrived migrants, for whom vaccination records are often paper-based, easily lost, or not integrated into national databases,¹⁹ undermining call and recall systems and continuity of care.

Administrative and procedural processes

Registration requirements such as identification numbers or translated records delay or prevent access to health care, including immunisation services, for refugees and undocumented migrants.^{19,22} Issuing multiple temporary identity documents (IDs) prevents linking health records

Panel 1: RIVER-EU, collaboration with Roma Health Mediators in Slovakia

In Slovakia, RIVER-EU addressed human papillomavirus (HPV) vaccination inequities among marginalised Roma communities by using and strengthening the Roma Health Mediator (RHM) system. RHMs, trusted community members embedded within Slovakia's national health system since the early 2000s, help to bridge gaps between Roma communities and health-care services, wherein mistrust, discrimination, and low health literacy impede access to care.³⁴

Through RIVER-EU, culturally adapted educational materials were co-created with RHMs, health-care professionals, and the National Healthy Regions programme to ensure both clinical accuracy and cultural resonance. These materials included PowerPoint presentations, brochures, and easy-to-understand leaflets to invite participation in educational sessions about HPV vaccination, translated into Romani and Slovak.³⁴

RHMs provide families with clear and culturally adapted information regarding the vaccine, its benefits, and how to access it, guiding them through the complex steps involved in the HPV vaccination process (eg, prescription and vaccine pickup). This approach is especially helpful for families with low health literacy or restricted mobility.³⁴

The intervention, implemented in 16 marginalised Roma communities, greatly improved HPV vaccination awareness and intent due to the trusted role of the RHMs and knowledge of the health system. Knowledge about HPV rose from 15% to 65% among Roma parents and from 30% to 70% among Roma schoolchildren. Intention to vaccinate increased significantly ($p < 0.0001$) in both groups. Among 192 parents who were followed up, 257 children eligible for vaccination were identified, and their HPV vaccination rates (20.6%) approached the regional average of 25.3%.

The collaboration between RIVER-EU partners and the Healthy Regions programme will facilitate educational sessions to continue as a part of health services and expand to other regions.

RIVER-EU exemplifies the essential role of professionals for health (building block 2). RHMs were a trusted source of information (2.1), expanded the capacity of the health system to reach the Roma community (2.2), received vaccine-communication training (2.3), and offered cultural representation (2.4). This active outreach (1.2), with tailored and translated information (1.3), was the result of a partnership between the Healthy Regions programme and RIVER-EU researchers (6.3) and supported trust in institutions (6.1).

across visits, leading to incomplete or duplicate courses.¹⁹ Even with electronic systems, complexity makes identifying unvaccinated individuals difficult,²¹ and many health systems do not have adequate protocols for managing missing records.¹⁹ Although tools such as Personal Child Health Records help track vaccinations and appointments, they are not universally issued.²⁷

Reminder and recall systems

Unified data systems, streamlined processes, and reminder systems have the potential to increase awareness of and appointment adherence for vaccination, especially for new users.²⁷ Automatic SMS reminders with appointment and rescheduling options are user friendly and reduce workload for PFHs.²⁰ However, sustaining tailored reminder systems incurs additional costs for already resource-constrained primary care teams.²⁷

Evaluation and monitoring

Evaluation and monitoring of vaccination programmes often does not have sufficient depth and coordination. Non-governmental organisation-led evaluations are rarely shared with state services. In AcToVax4NAM countries, professionals state that government monitoring usually counts doses administered rather than assessing coverage or barriers.¹⁹ Qualitative and disaggregated data could provide better insights into community experiences and inequalities.¹⁸ In prisons, the absence of interoperable systems forces continued reliance on paper records, thereby hindering data retrieval, obscuring immunisation status, and complicating continuity of care.³⁵

Building block 4: access to essential immunisation

Availability

In prisons and asylum centres, services are inconsistent and often outbreak-driven, and are further hampered by infrastructure challenges such as insufficient cold-chain storage.^{19,32,33} In Greek reception centres, vaccination has been described as chaotic and supply driven, resulting in ad-hoc delivery of whatever vaccines happened to be in supply at a given time (unpublished, RIVER-EU). Political debates about the so-called deservingness of asylum seekers or people in prisons also affect delivery.²⁹

Eligibility

Administrative barriers, such as temporary health IDs, delayed health cards, and strict age-based criteria can restrict access to vaccines. In Greece, provisional IDs restrict care for those with temporary status, with access revoked if asylum is denied.^{19,36} Conversely, consistent programmes build trust. In Israel, universal vaccine provision helped build perceptions of fairness despite broader societal discrimination.²⁰ In the UK and Israel, Bangladeshi and Arab parents trusted routine early childhood vaccines due to their long-standing, proven value and safety observed across generations and their role in eliminating diseases.^{19,27}

In prisons, offering vaccination aligned with national immunisation programmes fulfils the principle of equivalence of care.³⁷ Given the increased risk of vaccine-preventable diseases (VPDs) in prisons due to overcrowding, poor infrastructure, high prevalence of infectious diseases, and risk-related behaviours, an expanded vaccination offer could be considered.²⁵ Universal, gender-neutral HPV vaccination could be beneficial, given the elevated risk of

Panel 2: RISE-Vac, Moldova's large-scale campaign in prisons

Moldova's 17 prison facilities house more than 5700 individuals, with high rates of HIV, hepatitis C, and tuberculosis. Before the COVID-19 pandemic, no vaccination services were systematically available in these prisons. In March, 2021, Moldova launched its first large-scale prison vaccination campaign, led by the National Administration of Penitentiaries with the Ministry of Justice and supported by RISE-Vac.

The campaign aimed to expand equitable access to essential vaccines for people living and working in prisons and to align prison health services with national standards, notably the 2023–27 National Immunization Programme. The Ministry of Health provided vaccines free of charge, and prison health-care staff carried out vaccinations. All vaccinations administered were recorded in the national digital information system.³²

Initially, the vaccines administered included those against COVID-19, hepatitis B, influenza, and tetanus-diphtheria. At a later stage, the offer was extended to cover human papillomavirus (HPV) and measles, mumps, and rubella (MMR), all of which were managed under national cold-chain requirements. To implement the campaign, task shifting was applied; nurses took on organisational roles and custodial staff assisted in inviting people living in prison for vaccination. Educational activities, including face-to-face sessions, online information, and leaflets, were offered for health-care and custodial staff, in addition to people living in prison.³²

By Jan 1, 2024, 14 142 vaccine doses had been delivered, covering 2491 people living in prison for COVID-19, 1196 people living in prison for hepatitis B, and 2800 people living in prison for influenza. 1288 staff members were also fully vaccinated against COVID-19.³⁸ By June 30, 2025, vaccination coverage within people living in prison and prison staff had expanded further, with a total of 15 216 doses administered: 7867 for COVID-19, 3800 for hepatitis B, 2800 for influenza, 518 for diphtheria-tetanus, 131 for HPV, and 100 for MMR.

RISE-Vac shows the key role of access to essential immunisation (building block 4) through strong intersectoral commitment and policy alignment with the 2023–27 National Immunization Programme and the introduction of systematic vaccination services in prisons. The introduction of systematic vaccination services in prison, in turn, allowed for other components of health-system building blocks to function more effectively, including outreach and communication approaches (1.2), human resources capacity (2.2), training and confidence in communication and knowledge of rights (2.3), continuity and unification of data systems (3.1), and direct costs (5.1).

sexually transmitted infections, over-representation of key target groups (eg, sex workers), and high prevalence of sexual behaviours with high risk of infection.³⁵ Panel 2 includes a case study that highlights the role of access to essential immunisation.

Building block 5: financing

Direct costs

Vaccines not fully covered by national programmes tend to remain out of reach for underserved communities, including people living in prisons.³³ Even partial reimbursement rarely reduces this barrier,²² as additional payments are prohibitive for families already struggling with basic needs.¹⁴ By contrast, free vaccination programmes enable equitable access and build trust, as seen in the UK, where universally free childhood vaccines are widely valued.^{20,27}

Indirect costs

Travel expenses, lost income, and childcare needs can potentially deter vaccine uptake, especially for those in remote areas or with unstable employment.¹⁴ Additionally, digital barriers, such as a shortage of internet access or electronic devices for online booking, further hinder access.¹⁸ These indirect costs are closely associated with service delivery barriers (building block 1).

Systems funding

Underfunding of health systems affects vaccine procurement and delivery. In prisons, inadequate budgets restrict vaccine procurement and cold-chain maintenance.²⁹ In

resource-constrained settings, lower-cost vaccines such as measles, mumps, and rubella (MMR) are favoured over more expensive vaccines such as for HPV, leaving preventive needs unmet.¹⁹ Adapting vaccination schedules, such as accelerated hepatitis B virus or single-dose HPV regimens, can improve completion rates in prisons while releasing resources to expand coverage (eg, to older adults and to men).³⁵

Building block 6: leadership and governance

Trust and institutional frameworks

Trust in health institutions can heavily influence vaccine uptake.³⁹ In Finland, trusted public health nurses and child health clinics empower the vaccination decisions of parents.²⁶ However, trust is undermined when governance does not ensure inclusive and continuous services. The COVID-19 pandemic further strained trust in both institutions and vaccines, as health systems struggled to respond effectively and equitably and inconsistent communication fuelled confusion and scepticism.^{14,21} In prisons, distrust towards authorities persists likely due to underfunding, little guidance, and divided responsibilities between justice and health ministries.³³ Among undocumented migrants, fear of deportation, negative past experiences, and confusion about vaccination rights deter engagement.¹⁹

Guidelines and standard operating procedures

Only some European countries, such as Belgium, France, Latvia, Norway, and Slovakia, have national or subnational guidelines for prison vaccination, resulting in uneven

Panel 3: AcToVax4NAM, enhancing organisational vaccine literacy in first-reception centres in Italy

In Italy, the AcToVax4NAM project piloted a multicomponent intervention to transform vaccination delivery for newly arrived migrants who were hosted in first-reception centres or Centri di Accoglienza Straordinaria (CAS). The pilot conceptualised vaccination as a complex, multiphase process involving multiple stakeholders with differentiated roles and responsibilities both within the health-care system and at the community level.⁴⁰

The project, in close partnership with the local health authority and CAS staff, co-developed clear procedures to ensure that newly arrived migrants of all ages received vaccinations recommended in the National Immunisation Plan. Recognising that formal entitlements alone do not guarantee access to immunisation services, the Italian project team promoted organisational vaccine literacy.

The team launched a vaccination promotion campaign, producing easy-to-understand materials such as consent forms, screening questions, and post-vaccination advice sheets—all translated into nine languages commonly spoken by newly arrived migrants.⁴¹ The team also developed a glossary of essential vaccination terms⁴² written in clear, simple language for both health professionals and non-medical staff and delivered training to strengthen the CAS staff's knowledge of vaccine entitlements, schedules, and intercultural engagement.

As a result, vaccination rates among adult newly arrived migrants increased nearly 50-fold (vaccination rate was 7.54×100 person-month during the pilot vs 0.15×100 person-month in 2023) and overall coverage rose by 21.7% (7.60×100 person-month during the pilot vs 0.35×100 person-month in 2023). Health-care professionals reported significant gains in knowledge and confidence, with median knowledge scores rising from 1–3 points to 5 points (on a 5-point scale).⁴³

AcToVax4NAM highlights the value of building block component 6.2, guidelines and standard operating procedures, in facilitating improvements across other building blocks. These building blocks included components of service delivery (1.1, access and clinic hours, and 1.2, outreach and communication approaches), professionals for health (2.3, training and confidence in communication and knowledge of rights), and immunisation information systems (3.1, continuity and unification of data systems). The partnership between the local health authority and CAS staff (6.3) was also crucial.

service availability.³³ Absence of clear prioritisation leads to inconsistent COVID-19 vaccination approaches, despite the increased vulnerability of people living in prisons.²⁹ Even when migrants are included in immunisation plans, no explicit mention of them in policies can lead to their exclusion due to misunderstanding.¹⁹ The absence of standard operating procedures for vaccinating newly arrived migrants can lead to fragmented, inconsistent service delivery. Vaccinations are often administered opportunistically by non-governmental organisations rather than through systematic planning.¹⁹

Partnerships and integration

Building partnerships across sectors and mainstreaming vaccination into multiple contact points can help to strengthen reach. In the UK, consistent health promotion through trusted community institutions such as general practitioners' practices, health visitors, and mosques exemplifies effective integration at the governmental level.²⁹ In Israel, vaccinations delivered within school settings were identified as key to high vaccine uptake.²⁰ Integrating prison health within public health systems can improve service quality, staff capacities, and recognition of the rights of people living in prisons.³³ Collaboration between prison governance and health services through a whole-of-prison approach ensures continuity of care during detention and community re-entry.³⁰ For migrants, synergies between state services and civil society can help to address service gaps and improve referrals. Governance efforts that expand vaccination promotion to non-health settings such as labour offices, administrative offices, and other points of contact can further normalise immunisation and improve uptake (panel 3).¹⁹

Recommendations to overcome barriers and enhance enablers to vaccination for underserved populations

Building responsive immunisation programmes with constrained resources requires a realistic vision. Although no system can meet the needs of every individual, persistent barriers that disproportionately affect underserved populations are avoidable and continue to drive vaccine inequalities across Europe. Too often, interventions focus narrowly on individual behaviours or health literacy, while unwelcoming systems remain unchallenged. Structural reform and legislative action are essential, but meaningful progress is also possible through immediate service adaptations that increase accessibility and equity.

We offer five recommendations to address systemic barriers to equitable vaccination. These recommendations should be prioritised by policy makers and immunisation managers, alongside other enabling approaches we have described.

Human-rights approach to vaccination

Vaccination strategies for all populations should be grounded in fundamental human rights: the right to health, non-discrimination, and the state's duty of care for people under its responsibility. This approach aligns with international commitments to the World Health Assembly's resolution on universal health coverage,⁴⁴ Principle 16 of the European Pillar of Social Rights,⁴⁵ and the UN Standard Minimum Rules for the Treatment of Prisoners (Nelson Mandela Rules).³⁷

Immunisation policies should explicitly include people who are displaced, detained, undocumented, and other marginalised populations, not as an afterthought, but as a duty of care. Embedding a rights-based approach

strengthens equity and health-system integrity and ensures continuous, non-discriminatory access to vaccination across the life course. Structural barriers cut across age groups, from vaccination against MMR for children, against HPV for adolescents, and against COVID-19 for adults.

The WHO European Immunization Agenda 2030 prioritises vaccine equity and lifelong vaccination opportunities tailored to changing risks and needs.¹ All 53 member states within the WHO European region have adopted the Agenda and made commitments to achieve high immunisation coverage. In 2023, 45 (85%) of the member states reported having supportive legislation.³ Yet, suboptimal availability, barriers to access of vaccination services, and inconsistent and underdeveloped life-course strategies persist.

Standard guidance and tailored interventions

Even with supportive legislation and policies in place, the absence of clear protocols and accountability mechanisms for vaccination in settings where services are currently fragmented or ad hoc contributes to wide disparities in coverage. Standard operating procedures, applied across all phases of vaccination, can contribute to consistent and equitable delivery of immunisation. Clear guidance helps to assign responsibilities, ensures accountability, and improves monitoring and evaluation to track implementation, thereby supporting adaptive responses and evidence-informed policies.

Vaccination strategies need to be flexible and tailored to local contexts and needs, as the one-size-fits-all approach often overlooks specific cultural, linguistic, and logistical barriers. Tailored approaches, co-created through participatory research, are essential when addressing long-marginalised communities. The WHO Regional Office for Europe's Tailoring Immunization Programme offers a practical framework for diagnosing barriers and developing tailored solutions.⁴⁶ Within RIVER-EU, a methodological approach, participatory transferability analysis, has also been developed to tailor interventions. Participatory transferability analysis combines participatory action research with the Population-Intervention-Environment-Transfer Model of Transferability (PIET-T).^{47,48}

Decentralisation of services and intersectoral collaboration

Bringing vaccines into everyday settings where people live, study, work, and seek support removes barriers to access, supports a life-course approach to immunisation, and reconnects marginalised individuals with health care. Community-based immunisation services, delivered through schools, prisons, and reception centres, are effective when delivery models are tailored to local contexts and preferences. The 2024 EU Council recommendation on vaccine-preventable cancers calls for local provision of HPV and hepatitis B virus vaccines for groups at high risk,

including people living in prison, migrants, and people who are homeless.⁴⁹

However, decentralisation only works with strong intersectoral coordination. Health systems should collaborate with education services, social services, and civil society to ensure consistent messaging and culturally competent care. Such partnerships not only normalise vaccination within public services, but also build trust across community levels. They can also support addressing the information environment, as underserved populations are often disproportionately affected by misinformation and disinformation, exacerbated by digital algorithms that amplify polarising or misleading content.⁵⁰

Minimum dataset for vaccination

Increasing migration and population mobility across systems and regions make interoperable vaccination essential to avoid missed or duplicate doses; prison settings also experience highly mobile populations. A minimum dataset, coordinated across systems, would help track coverage and identify gaps, supporting continuity of care. This dataset should also actively inform policy and service planning, to improve access and equity. The European Health Data Space is a promising tool that explicitly includes vaccination information as part of its core minimum dataset for cross-border electronic health records.⁵¹ However, its full implementation remains years away and might exclude individuals with irregular migration status.

Immunisation data systems need to be designed with both the health system and individuals in mind. User-friendly tools, such as reminders, improve uptake among underserved populations.

Address underinvestment in health workforce and in health promotion and disease prevention

Health-workforce shortages risk deepening disparities, as underserved populations are often the first affected by limited service availability.⁵² Increasing investment in recruitment and training of PFH should prioritise diversity and cultural competence. Shared cultural backgrounds between providers and people can foster trust, facilitate communication, and boost vaccine uptake. The WHO European region's framework for action on the health and care workforce 2023–30 offers a clear roadmap for tackling these challenges.⁵³ RISE-Vac and AcToVax4NAM have also developed courses on vaccination, aimed at reaching professionals working with people living in prison and newly arrived migrants, which are available in multiple languages.^{54,55}

Simultaneously, health promotion and disease prevention remain underfunded, accounting for just 3% of the EU average share of health expenditure in 2019–20.⁵⁶ The EU Council 2024 conclusions on the Future of the European Health Union explicitly call for increased investment in these areas, to promote healthy, stigma-free environments that encourage vaccination alongside other prevention measures.⁵⁷ These conclusions further build on the

2018 recommendation from the EU Council on strengthened cooperation against VPDs, which, among other recommendations, encourages member states to support the training and skills development of health-care workers on VPDs, vaccinology, and immunisation.⁵

Conclusion

Information and experiences from RIVER-EU, AcToVax4NAM, and RISE-Vac show that tailored, community-centred approaches can effectively address long-standing structural barriers. Embedding equity into governance structures, strengthening intersectoral partnerships, and adapting services to people's lived realities can substantially improve access, trust, and vaccine uptake among underserved populations. When systems adapt to people instead of expecting people to adapt to systems, meaningful change becomes possible.

These learnings are transferable. They provide a valuable foundation of adaptable tools, trainings, guidance, strategies, and evidence for any immunisation programme aiming to tackle vaccine inequities for underserved populations. Crucially, they offer strong potential for policy exploitation at European, national, and regional levels. At a time when public trust in health systems is under pressure, a coherent, whole-of-government approach to immunisation is more urgent than ever. With sustained investments, inclusive policies, and accountability mechanisms, structural barriers can be dismantled and immunisation can be made the easy and obvious choice.

Contributors

AM, AN, JdZ, LT, PK, and ME conceptualised the paper and developed the methodology. AM and AN drafted the original manuscript. JdZ, LT, PK, and ME oversaw project administration. DFB, MET, MM, IB, and EDV drafted the panels. All authors contributed to the writing of the manuscript, reviewed and edited the manuscript, and had final responsibility for the decision to submit for publication.

Declaration of interests

This work was conducted as part of projects supported by EU funding: EU's Horizon 2020 programme (grant number 964353, RIVER-EU, SC1-BHC-33-2020) and cofunded by EU's Health Programme (2014–20; project number 101018353, RISE-Vac, HP-PJ-2020; project number 101018349, AcToVax4NAM, HP-PJ-2020). ME receives institutional grants from the Israel Science Foundation and the Israel National Institute for Health Policy Research and receives a research grant from MSD to investigate vaccination barriers to human papillomavirus among young adults in Israel. The content of this Health Policy reflects only the authors' views; the European Commission, the European Health and Digital Executive Agency, or any other body of the EU are not responsible for any use that could be made of the information in this Health Policy.

Acknowledgments

We thank the other members of the three project consortiums for their valuable collaboration and contributions throughout the course of this work. We are especially grateful to community representatives and health professionals who generously shared their experiences, insights, and time. Their input was the foundation of the findings presented in this Health Policy.

Editorial note: The Lancet Group takes a neutral position with respect to territorial claims in published text, tables, and institutional affiliations.

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